

CEN4802C Module 5 Project

For this project we will use GitHub Actions to automate the Docker components created in the Module 4 project for consistent building and testing.

- Start by adding the Module 4 Docker project code in your repo.
- Create the .github/workflows folder.
- Create a yaml file which checks out the code, configures the build, logs in to Docker hub, builds and runs the project, then stops it (see the sample workflow file, below).
- Add your Docker Hub login information to the repo secrets under Settings/Secrets and Variables/Actions (see "Setting Up Secrets", below).
- Add the yaml file to the workflows folder.
 - Each time you push a change, your workflow should execute.

Submit your repo to the GitHub classroom assignment and include screen snips demonstrating the operation of your workflow.

Setting Up Secrets

You need to set up Docker Hub credentials in your repo to allow the workflow to log in to Docker Hub.

- Click on Settings.
- Click on Secrets and Variables/Actions in the left sidebar.
- Click on New repository secret.
- Add two secrets: DOCKER_USERNAME and DOCKER_PASSWORD.

Sample Workflow File

```
name: Test Docker Compose

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

jobs:
  build:

    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Set up Docker Buildx
        uses: docker/setup-buildx-action@v3

      # set up for multi-platform docker
      - name: Set up QEMU
        uses: docker/setup-qemu-action@v3
```

- name: Log in to Docker Hub
uses: docker/login-action@v3
with:
 - username: \${ secrets.DOCKER_USERNAME }
 - password: \${ secrets.DOCKER_PASSWORD }
- name: Build and run Services
run: |
 - docker-compose -f docker-compose.yml up --build -d
 - sleep 15
 - curl --fail http://localhost:8080
 - curl --fail http://localhost:3000
- name: Stop Services
run: docker-compose -f docker-compose.yml down